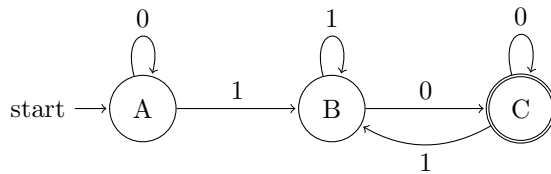


Rubrics Midsem CS310 2025

September 21, 2025

Q1

(a) 15 marks. Answer-



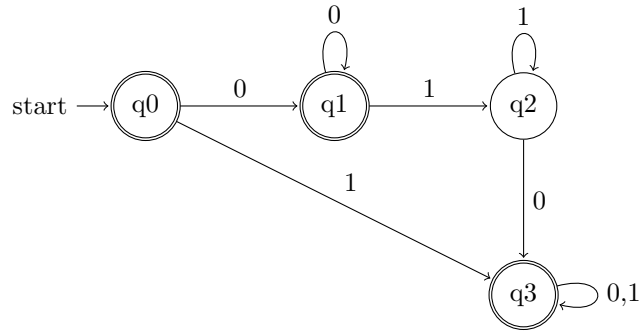
Score	Criteria
15	Correct answer
10	One wrong transition / Wrong final state
5	More than 3 states, with some mistakes
2.5	Reasonable attempt
0	Otherwise

(b) 5 marks. Answer- $(0 + 1)^*10^+$

Score	Criteria
5	Correct expression and length 3
3	Correct expression and length more than 3
1.5	Minor mistakes / Reasonable attempt
0	Otherwise

Q2

(a) 10 marks. Answer-



Score	Criteria
10	For correct DFA with 4 states
6	Almost a correct DFA (missing one transition or wrong label for at-most one state) with 4 states
2	Reasonable attempt
0	Otherwise

(b) 6 marks

Score	Criteria
6	For correctly showing the strings for every pair of states
4	For incorrectly/not showing the strings for at-most one pair of state
1	Reasonable attempt
0	Otherwise

(b) 4 marks. Answer- 122

Score	Criteria
4	for correct answer
0	Otherwise

Q3

(a) 2 marks.

Answer- A and B. 1 mark for each correct, -1 for each wrong, minimum 0.

(b) 4 marks

1 for each correct rule.

(c) 14 marks. One possible answer-

$$\begin{array}{l} S \rightarrow a|SS|SB|BS \\ B \rightarrow b \end{array}$$

Score	Criteria
14	Correct G_1 with ≤ 5 rules
10	Correct G_1 with 6 rules
8	Correct G_1 with ≤ 8 rules
6	Almost CNF having simplified to ≤ 3 variables and ≤ 7 rules or in CNF with more than 8 rules
2-4	Reasonable attempt
0	Otherwise

Q4

(a) 8 marks. Answer- L is not regular.

(b) 12 marks (minimum marks 0). Answer- There are many ways to justify/show L is not regular, but one possible way is to show that the complementary language is not regular using the pumping lemma by considering the words of the form 01^p0^{p-1} .

Case I: Using a closure property of the regular languages and then applying the pumping lemma:

Score	Criteria
12	Correct Justification with sufficient details.
-8	Incorrect application of the pumping lemma. Not mentioning the exact groups of words will also be considered an incorrect application.
-3	Missing to justify why the chosen language is regular if introducing any new language not obtainable from any of the operations that are closed under regular expressions, except trivial regexes
-3	Missing to state properly the required closure property of regular language.
-3	Not justifying why the pumping lemma fails.
2-4	Incorrect justification with reasonable attempt, but only if part-a is correct.
0	otherwise

Case II: Some other valid explanation:

Score	Criteria
12	Correct Justification with sufficient details.
6	Missing details (like making a correct assumption without justification)
2-4	Incorrect justification with reasonable attempt, but only if part-a is correct.
0	otherwise

These negative marks will be applied only if the overall justification is corrected. For example, even stated: "Take complement then apply pumping lemma", then it will receive $\max(12 - 8 - 3, 0) = 1$ marks.

Q5

(a) 5 marks. Answer- L is context-free.

Score	Criteria
5	Correct answer
0	Otherwise

(b) 15 marks

Score	Criteria
15	Correctly giving the CFG (≤ 7 rules) or PDA (≤ 5 states and 7 transitions)
10	Correctly giving the CFG (≤ 9 rules) or PDA (≤ 7 states and 8 transitions) along with proper explanation
4–7	Reasonable argument about L being regular
1–4	Arguing that L is irregular
0	Otherwise

Q6

(a) 8 marks.

Score	Criteria
8	Fully correct run
4	Correct run with skipped step(s)
0	otherwise

(b) 12 marks. Answer: $3^7 - 8$ or 2179

Score	Criteria
+6	For writing the correct value.
+5	For mentioning that given TM accepts word of the form $\omega\#\omega$
+1	Correctly justifying count after mentioning the above fact
+6	Other valid justification
-3	For missing details
2-4	Incorrect justification/value with reasonable attempt
0	otherwise